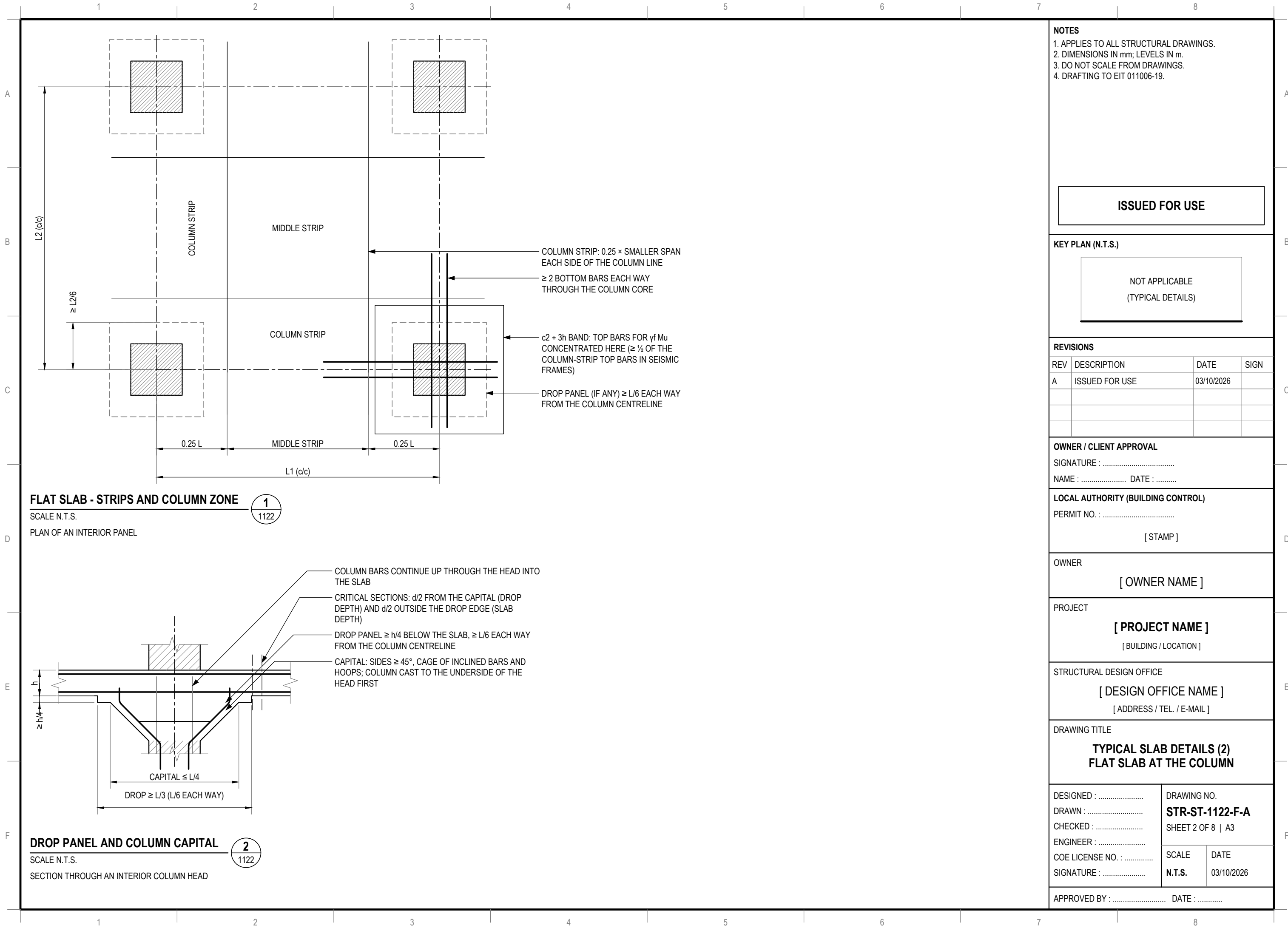
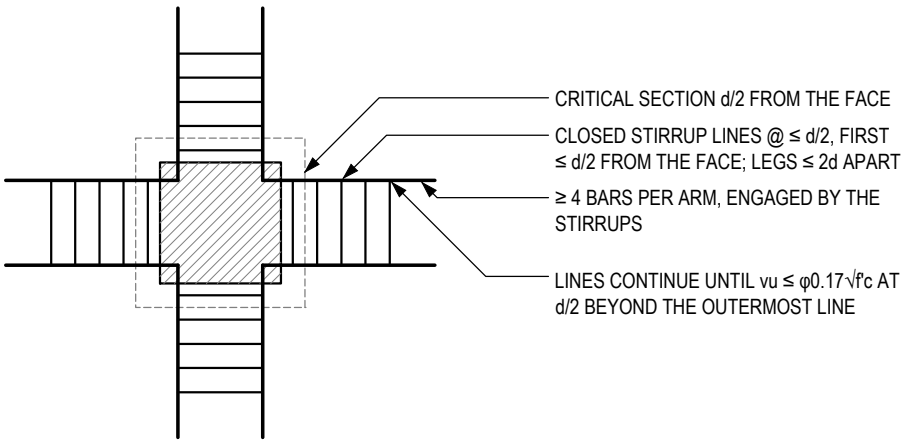




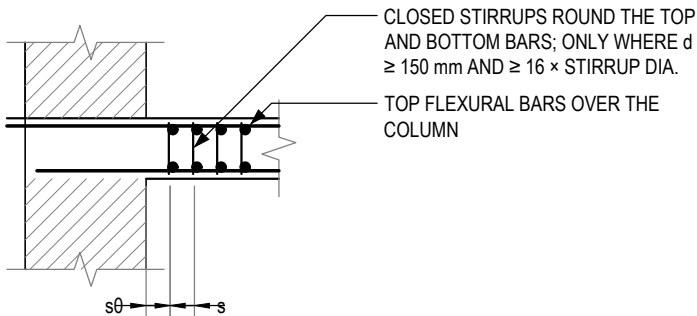


PUNCHING SHEAR - STIRRUPS

SCALE N.T.S.

PLAN AT AN INTERIOR COLUMN; RULES: TABLE 16

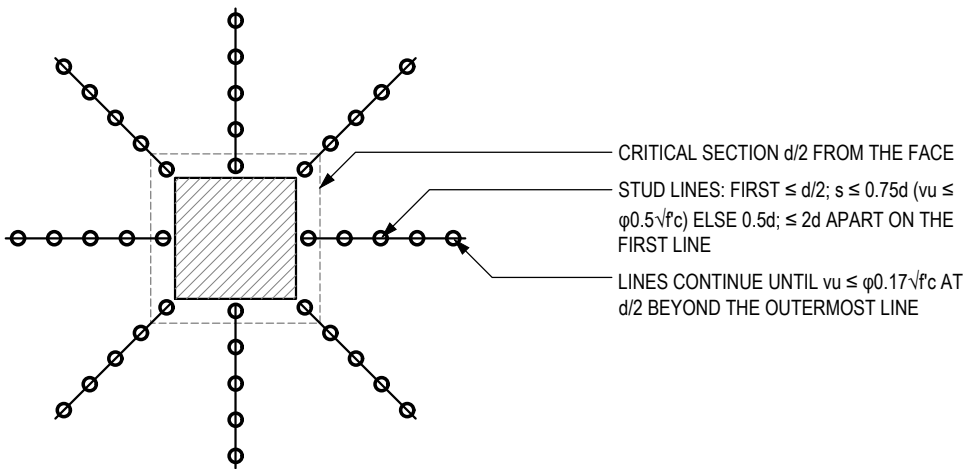
1
1123



SECTION

SCALE N.T.S.

A
1123

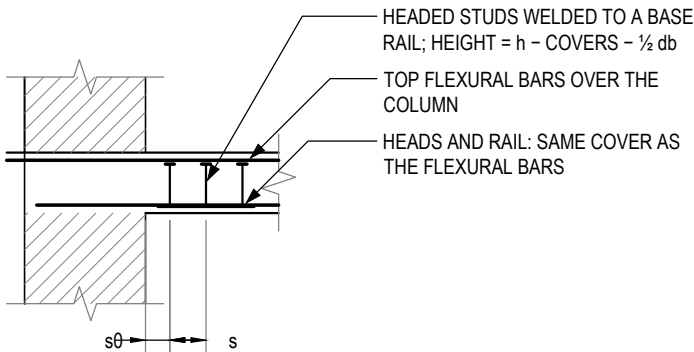


PUNCHING SHEAR - HEADED STUD RAILS

SCALE N.T.S.

PLAN AT AN INTERIOR COLUMN; RULES: TABLE 16

2
1123



SECTION

SCALE N.T.S.

B
1123

TABLE 16 - PUNCHING SHEAR REINFORCEMENT RULES

ITEM	STIRRUPS (BARS)	HEADED STUD RAILS
WHERE	$d \geq 150 \text{ mm AND } \geq 16 \times \text{STIRRUP DIA.}; \text{ CLOSED STIRRUPS ENGAGE THE FLEXURAL BARS}$	ANY SLAB DEPTH
STRENGTH LIMITS	$V_c \leq 0.17\sqrt{f_c} \text{ bo } d; V_n \leq 0.5\sqrt{f_c} \text{ bo } d$	$V_c \leq 0.25\sqrt{f_c} \text{ bo } d; V_n \leq 0.66\sqrt{f_c} \text{ bo } d; A_v f_{yt} / (b_o s) \geq 0.17\sqrt{f_c}$
FIRST LINE FROM THE FACE	$\leq d/2$	$\leq d/2$
SPACING OF LINES	$\leq d/2$	CONSTANT; $\leq 0.75d$ IF $v_u \leq 0.5\sqrt{f_c}$, ELSE $\leq 0.5d$
ALONG THE FIRST LINE	LEGS $\leq 2d$ APART	STUDS $\leq 2d$ APART
EXTENT	UNTIL $v_u \leq 0.17\sqrt{f_c}$ AT $d/2$ BEYOND THE LAST LINE	SAME
ANCHORAGE / COVER	STANDARD HOOKS ROUND LONGITUDINAL BARS (EIT 12.12)	HEIGHT = $h - \text{COVERS} - \frac{1}{2} db$; HEADS AND RAIL WITH THE FLEXURAL-BAR COVER
REFERENCE	EIT 011008 11.11.3	EIT 011008 11.11.5; ACI 318-11 7.7.5

NOTES

1. APPLIES TO ALL STRUCTURAL DRAWINGS.
2. DIMENSIONS IN mm; LEVELS IN m.
3. DO NOT SCALE FROM DRAWINGS.
4. DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

REV	DESCRIPTION	DATE	SIGN
A	ISSUED FOR USE	03/10/2026	

OWNER / CLIENT APPROVAL

SIGNATURE :

NAME : DATE :

LOCAL AUTHORITY (BUILDING CONTROL)

PERMIT NO. :

[STAMP]

OWNER

[OWNER NAME]

PROJECT

[PROJECT NAME]

[BUILDING / LOCATION]

STRUCTURAL DESIGN OFFICE

[DESIGN OFFICE NAME]

[ADDRESS / TEL. / E-MAIL]

DRAWING TITLE

**TYPICAL SLAB DETAILS (3)
PUNCHING SHEAR REINFORCEMENT**

DESIGNED :

DRAWN :

CHECKED :

ENGINEER :

COE LICENSE NO. :

SIGNATURE :

DRAWING NO.

STR-ST-1123-F-A

SHEET 3 OF 8 | A3

SCALE

N.T.S.

DATE

03/10/2026

APPROVED BY : DATE :

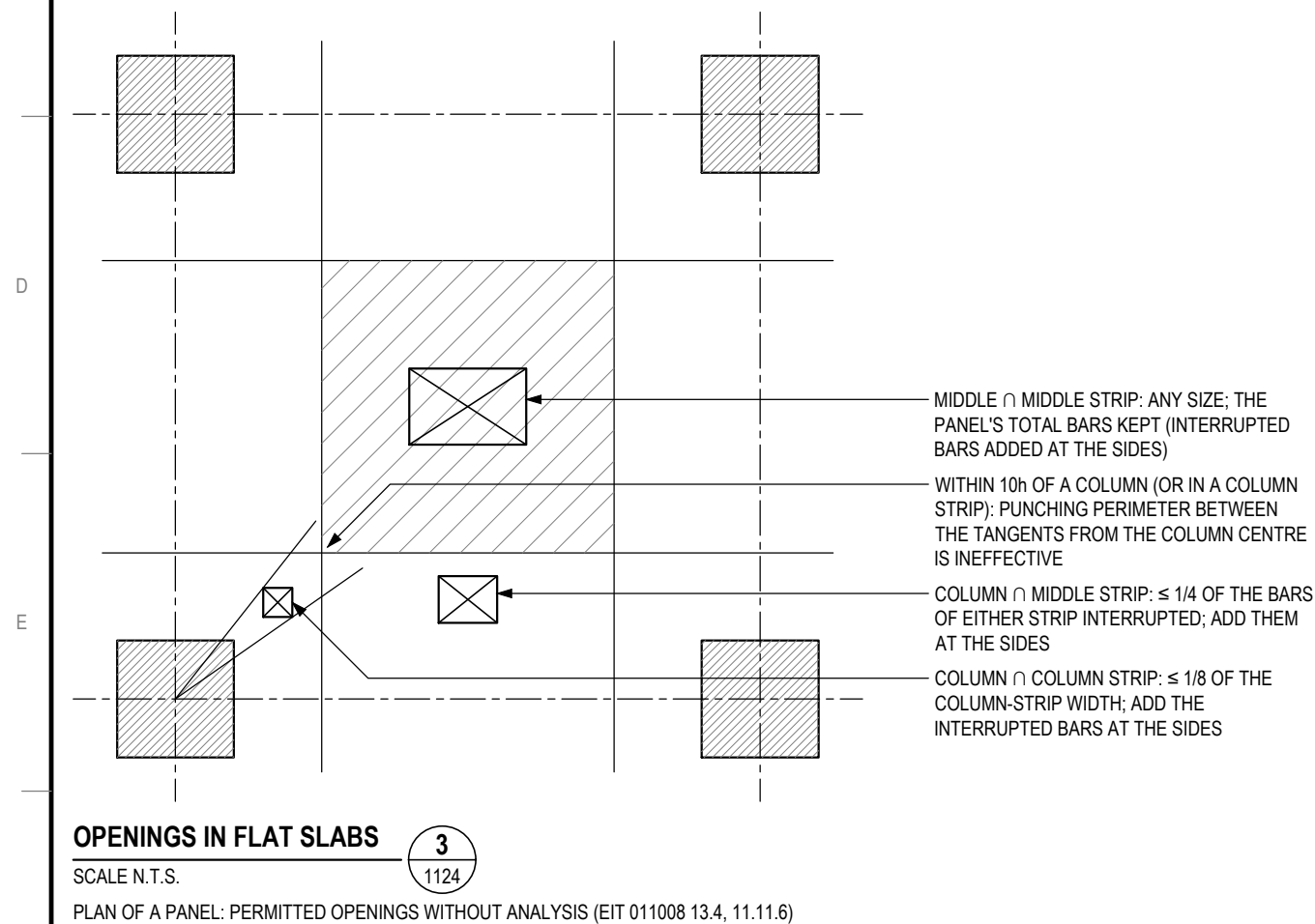
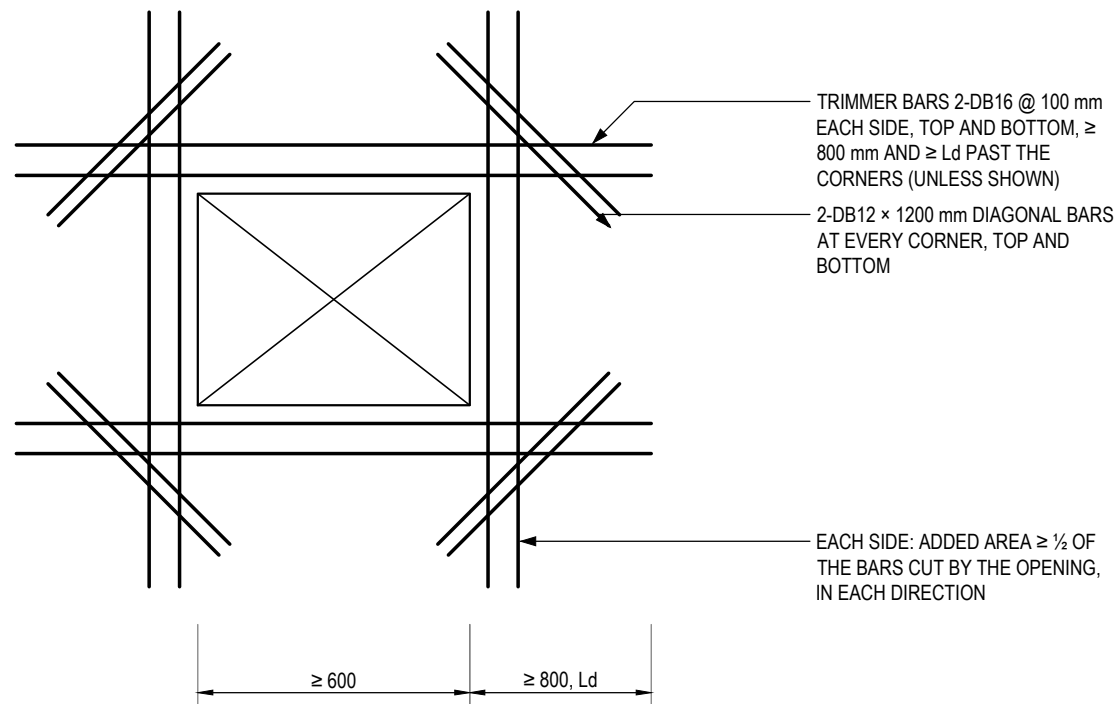
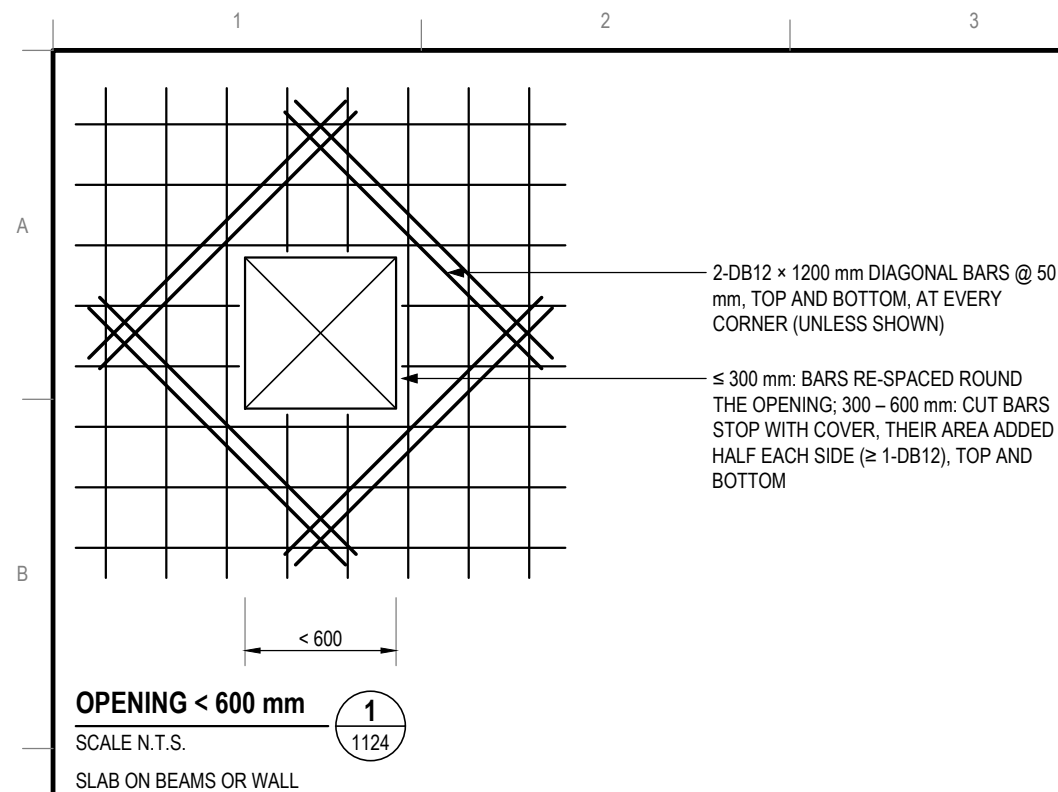


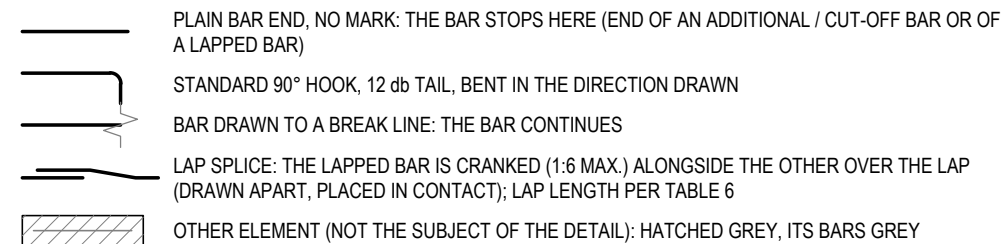
TABLE 17 - MINIMUM TRIMMING BARS AT SLAB OPENINGS

SLAB t (mm)	OPENING $\leq$ 300 mm	300 – 600 mm, EACH SIDE	$\geq$ 600 mm, EACH SIDE	DIAGONAL, EACH CORNER (L mm)
$\leq$ 125 mm (ONE MAT)	BARS RE-SPACED ROUND IT	1-DB12 CENTRAL	2-DB12 CENTRAL	1-DB12 $\times$ 1200
150 – 200	BARS RE-SPACED ROUND IT	1-DB12 T & B	2-DB16 T & B	2-DB12 $\times$ 1200 T & B
225 – 300	BARS RE-SPACED ROUND IT	2-DB12 T & B	2-DB20 T & B	2-DB12 $\times$ 1200 T & B

NOTES TO 1124

1. OPENINGS, SLEEVES AND CORED HOLES ONLY WHERE SHOWN ON THE STRUCTURAL DRAWINGS OR APPROVED BY THE ENGINEER; OPENINGS CUT LATER MUST BE TRIMMED TO RESTORE THE CUT STEEL.
2. BARS CUT BY AN OPENING STOP WITH COVER AT ITS EDGE (TOP BARS WITH A STANDARD HOOK DOWN); THEIR AREA IS ADDED BESIDE THE OPENING, HALF EACH SIDE, IN EACH DIRECTION, EXTENDED $\geq L_d$ PAST THE CORNERS, NOT LESS THAN TABLE 17. EXAMPLE: DB12 @ 150 mm, 800 mm OPENING: 5.3 BARS CUT = 6.0 cm² → 2-DB16 (4.0 cm²) EACH SIDE.
3. PRECAST HOLLOW-CORE PLANKS: HOLES $\leq \varnothing 150$ mm (OR ≤ 150 mm WIDE) ONLY THROUGH A CORE, ≤ 3 IN ONE CROSS-SECTION; LARGER OPENINGS ON STEEL-ANGLE TRIMMERS SEATED ON THE ADJACENT PLANKS.
4. OPENINGS THROUGH BEAMS: 1113. PIPES EMBEDDED IN SLABS: 1121 NOTE 5.

LEGEND



NOTES

1. APPLIES TO ALL STRUCTURAL DRAWINGS.
2. DIMENSIONS IN mm; LEVELS IN m.
3. DO NOT SCALE FROM DRAWINGS.
4. DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

REV	DESCRIPTION	DATE	SIGN
A	ISSUED FOR USE	03/10/2026	

OWNER / CLIENT APPROVAL

SIGNATURE :

NAME : DATE :

LOCAL AUTHORITY (BUILDING CONTROL)

PERMIT NO. :

[STAMP]

OWNER

[OWNER NAME]

PROJECT

[PROJECT NAME]

[BUILDING / LOCATION]

STRUCTURAL DESIGN OFFICE

[DESIGN OFFICE NAME]

[ADDRESS / TEL. / E-MAIL]

DRAWING TITLE

TYPICAL SLAB DETAILS (4) OPENINGS

DESIGNED :

DRAWN :

CHECKED : _____

ENGINEER ·

DRAWING NO.

STR-ST-1124-F-A

SHEET 4 OF 8 | A3

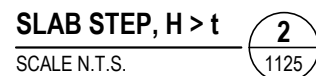
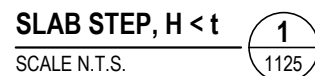
COE LICENSE NO. :

SIGNATURE:

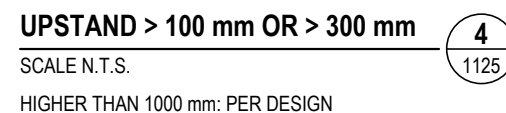
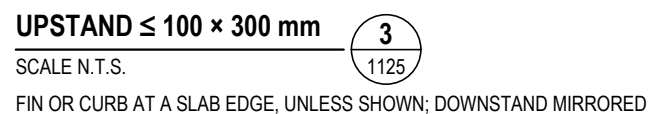
SCALE

NTC

APPROVED BY : DATE :



CANTILEVER SLAB 
SCALE N.T.S.
BALCONIES AND CANOPIES; LOADS PER DESIGN



APPROVED BY : DATE :

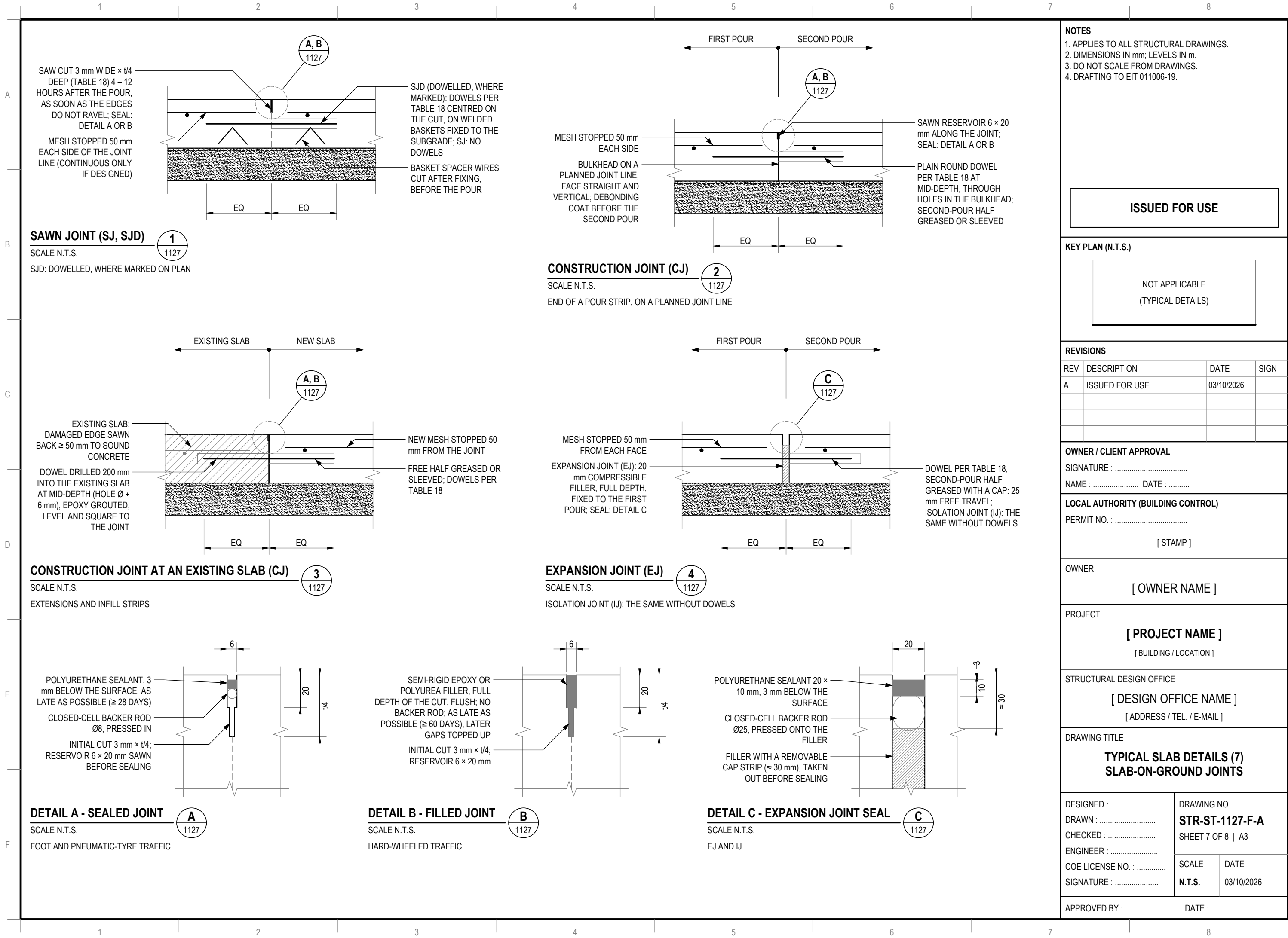
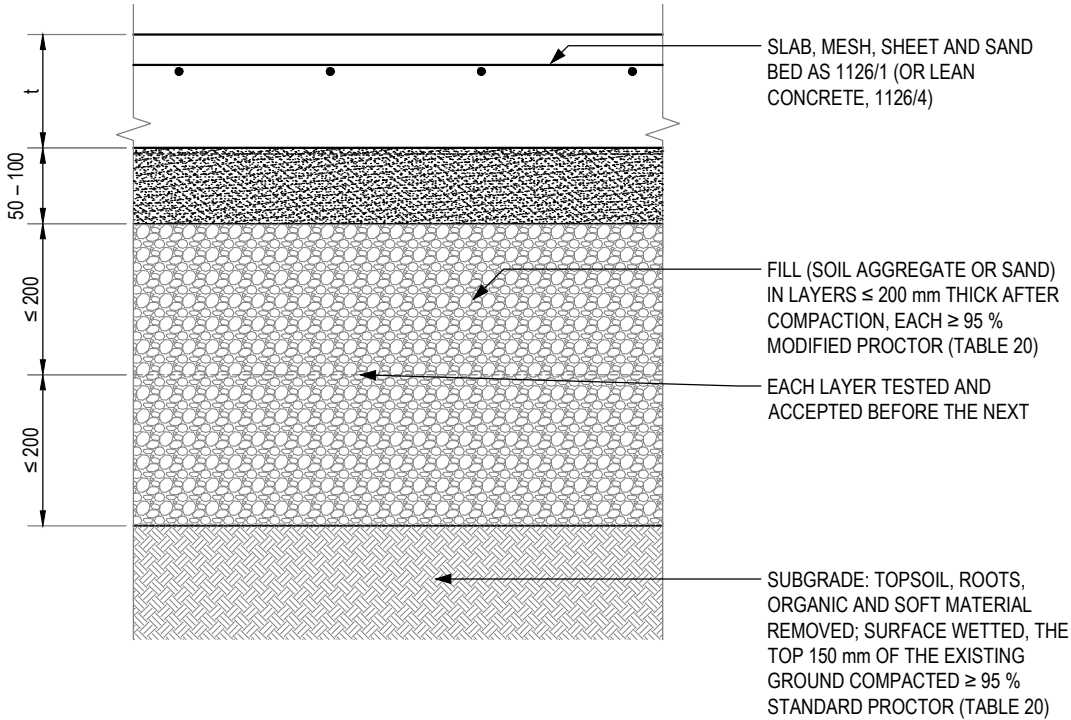


TABLE 19 - SLAB THICKNESS AND REINFORCEMENT

ITEM	ONE-WAY SLAB	TWO-WAY SLAB ON BEAMS	FLAT SLAB / FLAT PLATE
MIN. THICKNESS (NO DEFLECTION CHECK, SD40)	$\ell/20$ SIMPLE, $\ell/24$ ONE END CONT., $\ell/28$ BOTH CONT., $\ell/10$ CANTILEVER; $\times (0.4 + f_y/700)$ FOR OTHER STEEL	EQ. 9-11 / 9-12 ($\alpha f_m, \beta$); ≥ 125 mm ($\alpha f_m \leq 2$), ≥ 90 mm ($\alpha f_m > 2$); +10 % AT AN EDGE WITHOUT A STIFF BEAM	$\ell_n/30 - \ell_n/36$ (EIT 011008 TABLE 9.3); ≥ 125 mm, ≥ 100 mm WITH DROP PANELS
MAIN-BAR SPACING	$\leq 3h$ AND 450 mm	$\leq 2h$ AND 450 mm	$\leq 2h$ AND 450 mm
MIN. STEEL (EACH WAY, ON $b \times h$)	0.0025 SR24 / 0.0020 SD30 / 0.0018 SD40	SAME	SAME
DISTRIBUTION (S&T) BARS	SAME RATIO, PERPENDICULAR TO THE MAIN BARS, $\leq 5h$ AND 400 mm	–	–
TOP BARS AT SUPPORTS	$S_n/4$ AT EDGES, $S_n/3$ AT INTERIOR SUPPORTS	$S_n/4, S_n/3$ AND $L_n/4, L_n/3$	FIG. 13.3.8: 0.30 / 0.20 ℓ_n COLUMN STRIP, 0.22 ℓ_n MIDDLE STRIP
BOTTOM BARS	FULL SPAN, ≥ 150 mm INTO SUPPORTS	SAME; SHORT DIRECTION IN THE LOWER LAYER	COLUMN STRIP CONTINUOUS; ≥ 2 EACH WAY THROUGH THE COLUMN CORE
DISCONTINUOUS EDGE	TOP BARS 90° HOOK INTO THE EDGE BEAM	SAME + CORNER BARS $L/5 \times L/5$	TOP HOOKED, BOTTOM ≥ 150 mm; DEVELOP f_y AT THE FACE
REFERENCE	EIT 011008 7.6, 7.12, 9.5.2, 10.5.4, 12.10; TATA 3	EIT 011008 9.5.3, 13.3; TATA 3	EIT 011008 9.5.3, 13.3.8; DPT 5.2.12



SLAB ON COMPACTED FILL

SCALE N.T.S.
WHERE THE FLOOR IS RAISED ABOVE THE STRIPPED GROUND

TABLE 20 - SUBGRADE, FILL AND BACKFILL UNDER SLABS ON GROUND

LAYER	MATERIAL	PLACING	COMPACTION	FIELD DENSITY TESTS
SUBGRADE (EXISTING GROUND)	TOPSOIL, ROOTS, ORGANIC, SOFT AND UNSTABLE MATERIAL REMOVED (DPT 2112, 2115); ROOTS ≥ 300 mm BELOW THE UNDERSIDE OF THE WORK; HOLES REFILLED WITH FILL AND COMPACTED	SURFACE WETTED EVENLY; THE TOP 150 mm OF THE EXISTING GROUND COMPACTED. ON A SLOPE OR CUT: THE TOP 200 mm LOOSENED FIRST FOR BOND (DPT 2114)	≥ 95 % STANDARD PROCTOR (DPT 2201)	1 PER 500 m ² , ≥ 3 PER AREA
FILL: SOIL AGGREGATE / LATERITE (DEFAULT)	DPT 2101 4.2: MAX. SIZE 50 mm; ≤ 35 % PASSING 0.075 mm; CBR ≥ 8 % AT 95 % MODIFIED; SWELL ≤ 3 %; NO ROOTS OR ORGANIC MATTER	LAYERS ≤ 200 mm THICK AFTER COMPACTION; MIXED AND WATERED TO EVEN MOISTURE; EACH LAYER TESTED BEFORE THE NEXT	≥ 95 % MODIFIED PROCTOR (DPT 2202)	1 PER 500 m ² PER LAYER, ≥ 3
FILL: SAND	DPT 2101 4.3: NON-PLASTIC; MAX. SIZE 9.5 mm; ≤ 20 % PASSING 0.075 mm; CBR ≥ 10 % AT 95 % MODIFIED; NO CLAY LUMPS, TOPSOIL OR ORGANIC MATTER	LAYERS ≤ 200 mm THICK AFTER COMPACTION; ON SOFT GROUND NO VIBRATORY COMPACTION (NOTE 3)	≥ 95 % MODIFIED PROCTOR (DPT 2202)	1 PER 500 m ² PER LAYER, ≥ 3
FILL: SOIL (ONLY WHERE APPROVED)	DPT 2101 4.1: CBR ≥ 4 % AT 95 % STANDARD; SWELL ≤ 4 %; NO ROOTS OR ORGANIC MATTER	LAYERS ≤ 200 mm THICK AFTER COMPACTION	≥ 95 % STANDARD PROCTOR (DPT 2201)	1 PER 500 m ² PER LAYER, ≥ 3
BACKFILL AGAINST GROUND BEAMS, WALLS, PITS	AS THE FILL	LAYERS ≤ 200 mm THICK AFTER COMPACTION, SMALL PLATE OR RAMMER (METHOD APPROVED); BOTH SIDES OF A WALL AT ONE LEVEL (DPT 2114 5.1.7)	AS THE FILL	1 PER 50 m RUN PER LAYER
SAND BED UNDER THE SLAB	SAND AS ABOVE (DPT 2101 4.3)	50 – 100 mm THICK, MOISTENED, PLATE COMPACTOR	≥ 95 % MODIFIED PROCTOR (DPT 2202)	1 PER 500 m ²

NOTES
1. APPLIES TO ALL STRUCTURAL DRAWINGS.
2. DIMENSIONS IN mm; LEVELS IN m.
3. DO NOT SCALE FROM DRAWINGS.
4. DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

REV	DESCRIPTION	DATE	SIGN
A	ISSUED FOR USE	03/10/2026	

OWNER / CLIENT APPROVAL

SIGNATURE :
NAME : DATE :

LOCAL AUTHORITY (BUILDING CONTROL)

PERMIT NO. :
[STAMP]

OWNER

[OWNER NAME]

PROJECT

[PROJECT NAME]
[BUILDING / LOCATION]

STRUCTURAL DESIGN OFFICE

[DESIGN OFFICE NAME]
[ADDRESS / TEL. / E-MAIL]

DRAWING TITLE

TYPICAL SLAB DETAILS (8)
SLAB TABLE, FILL UNDER SLABS

DESIGNED :
DRAWN :
CHECKED :
ENGINEER :
COE LICENSE NO. :
SIGNATURE :

DRAWING NO.
STR-ST-1128-F-A
SHEET 8 OF 8 | A3
SCALE
N.T.S.
DATE
03/10/2026

APPROVED BY : DATE :